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1 October 1980
SUPERSEDING
MIL-G-52484A(ME)
26 January 1977

MILITARY SPECIFICATION

GRADER, ROAD, MOTORIZED, DIESEL-ENGINE-DRIVEN, SECTIONALIZED AND NON-SECTIONALIZED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air-droppable and air transportable heavy duty pneumatic-tired, diesel-engine-driven, 6-wheel, 4-wheel drive, front-wheel steer, articulated frame steer motor graders.

1.2 Classification. The graders shall be of the following types, as specified (see 6.2):

Type I - Non-sectionalized (Air Transporable and Air-droppable).
Type II - Sectionalized (Air Transportable, Air-droppable, and Helicopter Transportable).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-514	- Plates, Identification, Instructions and Marking, Blank.
MIL-G-762	- Graders, Road, Packaging of.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine Tactical Service.

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-M-7866	- Molybdenum Disulfide, Technical, Lubrication Grade.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-E-52798	- Enamel, Alkyd, Camouflage.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-669	- Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel.
MIL-STD-814	- Requirements for Tiedown, Suspension and Extraction Provisions on Military Materiel for Airdrop.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1410	- Methods for Selection of Industrial Engines For End Item Application.

MILITARY HANDBOOK

AIR FORCE SYSTEM COMMAND	
Design Handbook DH1-11	- Air Transportability.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).

TECHNICAL INFORMATION PACKAGE (TIP)

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HK, Fort Belvoir, VA 22060).

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3. REQUIREMENTS

3.1 General. The grader shall be the latest model of the standard product of the contractor and shall have demonstrated industry acceptability by having been manufactured, marketed, and sold in significant numbers to the commercial public for the same or similar application, interfacing with the same or similar components, except as specified herein, for at least 1 year prior to invitation for bids or request for proposal. Product improvements incorporated in the commercial production units in this 1 year period are acceptable. In the case of a recently introduced item, the contractor shall submit sufficient valid industrial operational/test data including all test reports for the item and its components, such as; engine, transmission and hydraulic system, field user reports, including any failures or problem areas, and corrective action taken to demonstrate acceptability of the item. When specified requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment, shall be furnished. The grader shall be equipped with any optional equipment necessary to comply with requirements specified herein. Optional equipment is defined as equipment not standard with the grader but which has been furnished to the commercial customer, such as special features or allied equipment. The grader shall be equipped with all components necessary to enable the grader to function reliably and efficiently in sustained operation. The grader shall conform to all Federal laws and regulations governing safety, noise levels, and pollution, which are in effect on the date of invitation for bids or request for proposal.

3.2 Description. The grader shall be pneumatic-tired, diesel-engine-driven, 6-wheel, 4-wheel drive, with front wheel steer, leaning front wheels, and articulated frame steer. The grader shall be equipped with a power shift transmission. A 12-foot moldboard and a V-type scarifer. The grader shall be operable by one man for general construction work such as course and fine grading, low and high bank sloping, flat and V-ditching, scarifying, and general construction and maintenance of roads. The grader shall permit true "sit down" operation with operator seated and seat belt properly fastened.

3.2.1 Type I. The type I grader with ROPS/FOPS removed shall be capable of being loaded and rigged on an air delivery platform in accordance with MIL-STD-669, air transported and air delivered by parachutes.

3.2.2 Type II. The type II grader with ROPS/FOPS removed shall meet requirements specified for type I (see 3.2.1) and shall also be capable of being sectionalized into 2 sections for external air transport by helicopter. Sectionalization material may be removed for airdrops.

3.3 First article (preproduction models). The contractor shall furnish two graders of each type for examination and demonstration within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce graders that comply with the requirements of this specification. The examination and demonstrations shall be as specified in Section 4. Following successful completion of the examination and demonstration by the contractor, the Government shall perform a simulated airdrop demonstration on one of each type grader at a site selected by the

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Government. The examination and all demonstrations conducted by the contractor shall be subject to surveillance and approval by the Government. Acceptance of the preproduction models will be dependent on successful completion of the examination, contractor demonstrations, and the Government conducted simulated airdrop demonstration.

3.4 Production graders. All graders furnished under this specification shall be physically and mechanically identical to the first article graders approved by the Government.

3.5 Material. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.6 Safety. All available standard or optional accessories shall be included to provide protection for maintenance and operating personnel from rotating or reciprocating parts and from all parts subject to high operational temperatures and so located as to be a hazard. Exhaust discharges shall be directed so as to not endanger personnel. All platforms and steps shall have anti-skid surfaces. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the grader can be set into motion. Contractor's standard grab irons shall be provided. A decal shall be furnished and shall be located in the area of the articulation point, warning of limited personnel clearance during frame articulation and placement of safety pin or bar for the grader operating and transport modes.

3.6.1 Sound level. The exterior sound level of the grader shall not exceed 88 dB(A) when measured in accordance with SAE J88. If the sound level in the area occupied by the operator exceeds 85 dB(A) when measured in accordance with SAE J919, a warning plate containing the following shall be permanently affixed to the grader in a conspicuous protected location.

"WARNING, EAR PROTECTION REQUIRED WHEN OPERATING THIS GRADER"

The plate shall have a yellow background with black lettering and shall be on corrosion-resistant material.

3.6.2 Rollover protective structure (ROPS) and falling objects protective structure (FOPS). The grader shall be equipped with a removable bolted down ROPS/FOPS canopy. The canopy shall be removable and replaceable without cutting or welding. The ROPS/FOPS shall conform to SAE J1040 and SAE J231 respectively. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. Quick disconnect fittings (electric, etc.) shall be provided as applicable to facilitate canopy removal. A sling eye(s) shall be provided to facilitate removal of ROPS/FOPS.

3.7 Physical characteristics.

3.7.1 Weight limits. The basic weight of the graders, type I and II without scarifier shall be not less than 28,000 pounds. Basic weight shall be determined with the grader in a fully serviced condition including fuel, oil,

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water, ROPS/FOPS, and with standard accessories as specified herein. Basic weight shall not include the scarifier, optional items, hydroflation or other tire ballast, bolt-on wheel weights or other bolted-on or weld-on weight, or an operator. For airdrop, the grader without the ROPS/FOPS, without scarifier, and without sectionalization kit, but with all other attachments, accessories, coolant, lubricating and hydraulic systems filled to capacity and with not less than 25 percent nor more than 75 percent fuel shall weigh no more than 30,600 pounds. The type II grader when sectionalized into 2 sections shall weigh not more than 16,000 pounds per section.

3.7.2 Dimensions. The overall width of the grader shall not exceed 98 inches, except that the width may be increased to not more than 105 inches to accomodate special airborne/airmobile provisions. The overall length including scarifer shall not exceed 361 inches. In order to conform to a height limit of 100 inches when rigged for airdrop, the reducible height of the grader should be not more than 91 inches. No major component removal except ROPS/FOPS canopy, shall be required to obtain the reduced height. If the height exceeds 91 inches, the contractor must demonstrate prior to step I proposal opening date that his grader can be rigged so as to conform to the 100 inch limit, and withstand the forces encountered in the simulated airdrop impact test (see 4.5.2.12) without damage or permanent deformation.

3.8 Sectionalization. The type II grader less ROPS, shall be capable of being sectionalized into two Sections for helicopter transport. The weight of either section shall not exceed 16,000 pounds. The powered section shall be driveable and steerable under its own power for a distance of 50 feet over relatively flat unimproved areas. Reassembly time with tools furnished with the grader under field conditions without assistance of other powered equipment shall not exceed 4 hours. Cutting and welding are not permitted. Components and accessories used in sectionalization and reassembly shall be stored on the grader or in a demountable container attached to the grader but shall not interfere with the operation of the grader.

3.9 Performance. The grader and attachments shall withstand the useage encountered in military operations such as rough and fine grading, ditching, high bank cutting, and sloping, grading in reverse and scarifying. The grader shall withstand crossing roadless rough terrain, including gullies causing the tandems to oscillate in both directions, fording streams, and negotiating steep lontitudinal and sidehill slopes, without damage. The grader shall be capable of driving at maximum travel speed in transport, being towed by another vehicle, and limited towing of other equipment.

3.9.1 Environmental conditions. Without the use of any external means, equipped with only standard and optional equipment, the engine shall start within five (5) minutes and the grader shall operate under full-load conditons within fifteen (15) minutes after engine start in any ambient temperature from minus 25° F to plus 120° F. Cold start aids must be intergal with the grader and no aids may be used which rely upon external support such as an electrical power source. Hand-held aerosol cans are not permitted. The grader shall be capable of operating in hot humid tropical environments and dry dusty environments.

3.9.2 Travel speed. The grader shall be capable of a forward travel speed of not less than 20 mph on level pavement.

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3.9.3 Blade pull. The grader shall develope a blade pull of not less than 16,000 pounds at 2 plus or minus 1/2 mph, at the specified operating weight, on a dry, level, concrete surface, with no more than 15 percent slip factor.

3.9.4 Stability. The grader shall negotiate 35 percent side slopes and 60 percent longitudinal slopes without any wheel of the grader losing contact with the ground. All parts requiring lubrication shall be lubricated when the grader is operating under these conditions.

3.9.5 Fording. The grader shall start and operate during and after fording operations in fresh water to any depth up to 20 inches including wave action without damage and without leakage of water into reservoirs containing lubricants.

3.10 Engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The diesel engine shall be the engine furnished commercially including all systems, components, accessories, and auxiliaries. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow.

3.10.1 Engine horsepower. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816 shall be not less than 135 horsepower.

3.10.2 Engine cooling system. The contractor's standard or optional cooling system meeting the requirements of 3.9.1 shall be provided.

3.10.3 Fuel tank. The fuel tank shall be of a type regularly furnished as standard equipment with a capacity for 10 hours of continuous normal operation. The tank shall be equipped with a means for draining water and sediment from the bottom of the tank.

3.10.4 Engine side panels. The grader shall be equipped with removable engine side panels with quick-disconnect latches or horizontally or vertically hinged engine compartment doors.

3.11 Drive train.

3.11.1 Transmission. The grader shall be equipped with a power shift transmission capable of being shifted under full engine power through all successive forward and reverse gear ratios without deceleration of the grader. Not less than four forward speeds and four reverse speeds shall be provided in the transmission. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the grader can be set into motion.

3.11.2 Torque converter. Graders equipped with a torque converter shall have gear reduction suitable to keep the torque converter working efficiently throughout the entire range of grader speeds.

3.11.3 Tandems and differential. Tandem oscillation shall be not less than 18 degrees from the horizontal in either direction. Tandem oscillation shall be

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limited to prevent excessive rotation when the grader is being lifted using the lifting eyes or suspension provisons. The grader shall have a manual or automatic lock-unlock drive train differential.

3.12 Wheels and tires. Commercial standard demountable wheels or rims shall be provided. Traction type tubless tires shall be furnished and shall have a tread pattern suitable for grader operation. Tire size shall be not less than 13 x 24 with not less than 10-ply rating. All tires shall be the same size, type and ply rating.

3.13 Steering and braking.

3.13.1 Steering. The grader shall have hydraulically boosted mechanical or full hydraulic power steering, and articulated frame steering. The steering system shall provide full control of the grader under all normal operating conditions and shall provide emergency steering conforming to SAE J53 without engine power. An articulation position indicator shall be furnished. The leaning front wheels shall be capable of leaning at least 15 degrees right and left.

3.13.2 Braking system. A foot controlled self-adjusting sealed disc service braking system shall be provided and shall be effective on the four driving wheels. A parking brake system and an emergency stopping system shall be provided. The braking systems shall comply with SAE J1152.

3.14 Operator's compartment.

3.14.1 Seat. The contractor's standard commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, and visibility of the work zone, and all instruments and gages mounted in the operator's compartment, when seated with seat belt fastened and properly adjusted.

3.14.2 Instruments and indicators. The following instruments and indicators shall be provided:

(a) Engine oil pressure.
(b) Engine coolant temperature.
(c) Transmission/torque converter oil temperature (if offered).
(d) Ammeter or voltmeter.
(e) Engine hourmeter or service meter.
(f) Air pressure gage (when equiped with air system).
(g) Low air pressure warning buzzer (when equipped with air system).
(h) Fuel level indicator (gage or dipstick).
(i) Hydraulic oil level indicator (dipstick, sight tube or gage) (if offered).
(j) Air cleaner restriction gage.
(k) Odometer (if offered).
(l) Speedometer or tachometer (if offered).

Additional instruments and indicators standard with the contractor's commercial practice, which contribute to the safety and reliability of grader operation, shall be furnished.

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3.14.3 Controls. All controls for the grader and attachments shall be located in the operator's compartment and shall be within reach of the operator in his normal operating position.

3.15 Hydraulic system. The grader shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all attachments and related components. All movements of the blade, scarifer and leaning wheel mechanisms shall be controlled through hydraulic valves.

3.16 Electrical system. The contractor's standard or optional 12 or 24 volt electrical system shall be furnished. The starting system shall be 24 volts if available as standard or as an option.

3.16.1 Battery. The battery or batteries shall be the contractor's standard or optional batteries for cold weather starting as specified in 3.9.1.

3.16.2 Alternator. The contractor's high output option alternator shall be furnished. The rated output shall be not less than 50 amps.

3.16.3 Lighting. The following lights with associated circuits, circuit breakers, or other protective devices, and switches shall be provided if available as standard or as an option:

(a) Two headlights.
(b) Two front working lights.
(c) Combination stop and taillights.
(d) Turn signals with emergency flasher.
(e) Rear floodlight(s).
(f) Instrument panel lighting.

3.16.4 Electromagentic radiation. The grader shall comply with the electromagnetic radiation limits of SAE J551.

3.16.5 Horn. A conventional operator controlled horn shall be furnished.

3.16.6 Back-up alarm. A back-up alarm conforming to SAE J994 shall be furnished.

3.17 Towing devices.

3.17.1 Front tow hook. A tow hook or other suitable means of attaching a tow cable or tow bar shall be provided on the front of the grader.

3.17.2 Rear tow hitch. The grader shall be equipped with a rear tow hitch including a pin and a positive pin retaining means.

3.18 Attachments.

3.18.1 Blade. The blade assembly shall consist of a wear-resistant moldboard and replaceable cutting edges and shall have hydraulic sideshift and hydraulic tilt. The blade shall be not less than 12 feet long, 2 feet high, and 3/4 inch thick.

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3.18.1.1 Blade lift. The blade shall have a lift above ground level of not less than 16 inches.

3.18.1.2 Blade cut. The blade shall have a cut below ground of not less than 10 inches.

3.18.1.3 Blade reach. The blade shall have a reach (right/left) of not less than 6 feet outside the rear tires.

3.18.1.4 Bank cutting angle. The blade shall have a right/left bank cutting angle of 90 degrees.

3.18.1.5 Circle turn. The blade shall be capable of traversing 360 degrees without removing any machine components other than scarifier shanks by utilizing controls at the operators station.

3.18.2 Scarifier. The grader shall be equipped with a V-type scarifier. The scarifier shall be equipped with removable shanks having replaceable points. The scarifier shall have not less than 9 shanks and a swath of not less than 46 inches.

3.19 Transportability. The grader shall be capable of being transported by surface, rail, ocean-going carriers, and C130, C141, and C5A aircraft. The grader shall withstand the impact forces encountered in air and rail shipments without damage or permanent deformation. The grader shall conform to MIL-STD-669 for air drop from C130 aircraft and shall withstand stresses imposed when tested in accordance with 4.5.2.12 without breakage or permanent deformation of any structural member or component, or loss of ability of the grader to perform all of its intended functions. When rigged for airdrop in accordance with MIL-STD-669, the grader shall meet the established tipoff curves for C130 and C141 aircraft as defined in AFSC Design Handbook DH1-11. The grader shall be equipped with slinging provisions and shall be capable of being lifted without damage using slings. The grader shall be equipped with tiedown provisions for transport by the above modes. Suspension and extraction parachute provisions shall be provided for parachute delivery. Complete diagrams and instructions for lifting the grader and tying down the grader on freight cars and truck trailers shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport. The contractor shall provide sufficient data to the Government with the step I proposal to confirm that the grader being supplied can be loaded and transported in the C130, C141 and C5A aircraft. A steering frame lock in accordance with SAE J276 shall be provided. Removal of ROPS and other components to reduce overall height or width shall not prevent the grader from driving onto military transport equipment under its own power.

3.19.1 Parachute suspension provisions. Four integral suspension provisions conforming to MIL-STD-814 shall be furnished on the grader. The parachute suspension provisions shall withstand the stresses in the amount and direction of pull specified without weld failure or permanent deformation of the grader or the provision.

3.19.2 Slinging provisions. The grader shall be provided with four slinging provisions conforming to MIL-STD-209. In addition, each section of the type II grader shall be provided with four slinging provisions conforming to

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MIL-STD-209. The provisions shall enable the grader to be lifted in the normal traveling or operating position. Provisions shall be positioned so that the lifting strain shall be in line with the longitudinal axis of the eye of the provision. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provision without weld failure or permanent deformation of the grader or the provision. Parachute suspension provisions may be used as slinging provisions if they meet the requirements of MIL-STD-209.

3.19.3 Tiedown provisions. The grader shall be provided with integral provisions to permit tiedown of the grader to the floor or deck of the transportation medium or the airdrop platform. The tiedown provisions shall conform to MIL-STD-209 and MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the grader or the provision.

3.19.4 Extraction parachute provision. A single fitting shall be provided on the grader for attachment of the parachute extraction system. Except as specified herein, the fitting shall conform to MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the grader or the provision. In lieu of the limit load specified in MIL-STD-814, the limit load of the provision shall be 56,580 pounds. The design ultimate strength of welded sections shall be four times the limit load.

3.20 Identification plates and marking. Each grader shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, serial number, weight, date of manufacture and date of delivery. The identification plate shall be permanently affixed to the grader in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. All attachments or components removed or disassembled for shipment shall be matchmarked for proper reassembly. The grader and each attachment shall be identified in accordance with the contractor's standard method.

3.21 Instruction plates. The grader shall be equipped with instruction plates including warnings and cautions suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the grader. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. Instructions for disassembly of the type II grader into two sections and reassembly shall be provided on cards approximately 8-1/2 inches by 11 inches which shall be encased in laminated plastic. One set shall be furnished with each type II grader.

3.22 Shipping data plates. Shipping data plates for each grader shall be furnished and shall show the silhouette of the grader and each section (type II grader only), indicating the location and capacity of tiedown provisions, slinging provisions, extraction provisions, suspension provisions, the center of balance and shipping weight. The plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. The plates shall be attached to the left side of the grader and shall show the

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locations as viewed from the left side. The plates on the sections, (type II grader only) shall be located in conspicuous protected locations. All plates shall be permanently attached.

3.23 Treatment and painting. The portions of the grader normally painted shall be cleaned, treated, primed and painted in accordance with the contractor's standard commercial practice. The finishing coat shall be in accordance with MIL-E-52798, forest green color.

3.24 Stenciling instructions. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown, slinging, suspension, and extraction provisions, tire pressure and the center of gravity for the type I and type II grader, and the sectionalization storage areas on the type II grader.

3.25 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each grader shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924, and MIL-G-23827 as applicable.

3.26 Technical publications. Such technical publications as are specified (see 6.2) shall be furnished.

3.27 Special tools. Any nonstandard or special tools required for equipment operation and operator maintenance of the grader shall be furnished with the grader.

3.28 Toolbox. A lockable storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided.

3.29 Technical information package. The contractor shall complete and submit a Technical Information Package within the time frame specified (see 6.2), depicting the grader and its components exactly as it will be furnished. The Technical Information Package shall be submitted with all bids and proposals.

3.30 Government-furnished equipment (GFE). The following items shall be furnished by the Government (see 6.3):

(a) Container NSN 7520-00-559-9618 complete with maintenance forms. (The contractor shall furnish the fasteners and install the containers on the grader.)
(b) Supplemental operating, maintenance and repair parts instructions. (SOMARPI).

3.31 Basic issue items. When specified (see 6.2), each item in the Basic Issue Item List shall be furnished with each grader.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

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contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection. The first article (preproduction model) graders (see 6.2), furnished by the contractor shall be examined (see 4.3.1), and demonstrated (see 4.3.2), to determine compliance with the requirements of this specification. Prior to the examination and demonstration of the graders, the following shall be performed:

(a) Service the graders with the lubricants specified.
(b) Break in the graders as prescribed by the contractor.

4.3.1 Preproduction examination. Prior to the preproduction demonstration, the graders shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection. Prior to the preproduction demonstration the graders shall be examined and compared with this specification, with the contractor's published brochures, with commercial manuals, and with the completed Technical Information Package. Any discrepancy between the specification, brochures, manuals, or Technical Information Package and the preproduction models shall be cause for rejection. Prior to the preproduction demonstration, the certifications, and certified test reports required by 4.5.2.13 and 4.5.2.14 shall be examined. Incomplete, inaccurate, or otherwise unacceptable certifications, or certified test reports shall be cause for rejection.

4.3.2 Preproduction demonstration. Upon successful completion of the examination specified in 4.3.1, the preproduction model graders shall be subjected to the demonstrations marked "X" in column 1 of table II. Failure of any demonstration or any demonstration reports shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the demonstration specified in 4.4.2 each grader shall be examined for the defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual demonstration. Each grader shall be demonstrated as specified in column 2 of table II. Failure of any demonstration shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with table I.

4.5.2 Demonstrations. Demonstrations shall be conducted in accordance with table II in the sequence listed.

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TABLE I. Examination schedule.

Preproduction Model(s)	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	101. Grader type not as specified.	1.2
X	X	102. Grader description not as specified.	3.2
X	X	103. Safety characteristics not as specified.	3.6
X	X	104. Articulation warning decal not furnished.	3.6
X	X	105. Sound level warning plate not as specified.	3.6.1
X	X	106. ROPS/FOPS not furnished or equipped with disconnects and lift eye.	3.6.2
X	-	107. Dimensions not as specified.	3.7.2
X	X	108. Sectionalization and components not as specified.	3.8
X	X	109. Engine not as specified.	3.10
X	X	110. Fuel tank not as specified.	3.10.3
X	X	111. Engine side panels not as specified.	3.10.4
X	X	112. Transmission and torque converter not as specified.	3.11.1, 3.11.2
X	-	113. Tandem oscillation not as specified.	3.11.3
X	X	114. Differential not as specified.	3.11.3
X	X	115. Wheels and tires not as specified.	3.12
X	X	116. Steering not as specified.	3.13.1
X	X	117. Articulation position indicator not furnished.	3.13.1
X	-	118. Front wheel lean not as specified.	3.13.1
X	X	119. Braking system not as specified.	3.13.2
X	X	120. Seat not as specified.	3.14.1
X	X	121. Instruments not as specified.	3.14.2

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TABLE I. Examination schedule. (Cont'd)

Preproduction Model(s)	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	122. Hydraulic system not as specified.	3.15
X	X	123. Electrical system not as specified.	3.16
X	X	124. Batteries not as specified.	3.16.1
X	X	125. Alternator not as specified.	3.16.2
X	X	126. Lighting not as specified.	3.16.3
X	X	127. Horn not as specified.	3.16.5
X	X	128. Back-up alarm not as specified.	3.16.6
X	X	129. Front tow hook not as specified.	3.17.1
X	X	130. Rear tow hitch not as specified.	3.17.2
X	X	131. Blade not as specified.	3.18.1, 3.18.1.1, 3.18.1.2, 3.18.1.3, 3.18.1.4, 3.18.1.5
X	X	132. Scarifier not as specified.	3.18.2
X	-	133. Transportability not as specified.	3.19
X	X	134. Identification plates not as specified.	3.20
X	X	135. Instruction plates not as specified.	3.21
X	X	136. Shipping data plates not as specified.	3.22
X	X	137. Treatment and painting not as specified.	3.23
X	X	138. Stenciling instructions not furnished.	3.24
X	X	139. Lubrication not as specified.	3.25
X	X	140. Technical publications not as specified.	3.26
X	X	141. Special tools not furnished.	3.27
X	X	142. Toolbox not furnished.	3.28
X	-	143. Technical information package not as specified.	3.29

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TABLE I. Examination schedule. (Cont'd)

Preproduction Model(s)	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	144. Government furnished containers not installed on grader.	3.30
X	X	145. Basic issue items (if specified) not furnished.	3.31
X	X	146. Parts or components missing.	Section 3
X	X	147. Assembly incomplete or incorrect.	Section 3
X	X	148. Cracked or broken parts or assemblies.	Section 3
X	X	149. Permanent deformation of any part or assembly.	Section 3
X	X	150. Leaks in coolant, fuel, oil lines, or cylinders.	Section 3
-	X	151. Material not of same quality used on preproduction graders.	Section 3
X	X	152. Grader not as specified.	Section 3

TABLE II. Demonstration schedule.

Preproduction Models		Individual	Test	Test Paragraph	Requirement Paragraph
1		2	3	4	5
Type I	Type II				
X	X	X	Production run-in	4.5.2.1	Section 3
-	X	-	Sound level	4.5.2.2	3.6.1
X	X	-	Weight	4.5.2.3	3.7.1
X	X	-	ROPS removal	4.5.2.4	3.6.2 and 3.19
-	X	-	Sectionalization	4.5.2.5	3.8
X	X	-	Parachute suspension provisions	4.5.2.6	3.19.1
X	X	-	Slinging provisions	4.5.2.7	3.19.2
X	X	-	Tiedown provisions	4.5.2.8	3.19.3
X	X	-	Extraction provisions	4.5.2.9	3.19.4
-	X	-	Construction tasks	4.5.2.10	3.9 and 3.10.3
-	X	-	Blade pull	4.5.2.11	3.9.3
X	X	-	Airdrop	4.5.2.12	3.19

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4.5.2.1 Production run-in. Start and operate each grader as follows:

Operate the grader in all gears forward and reverse at maximum speeds commensurate with conditions for one-half hour. Stop and restart the engine not less than five times. Operate all blade and scarifier controls for five-minute periods. Hold each hydraulic control in each actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the grader through not less than 10 full right and 10 full left turns using maximum wheel lean at the maximum speed commensurate with safe operation. Make not less than five abrupt panic stops from a speed not less than 20 MPH using the service brake. Release and reapply the parking brake not less than five times. Turn on all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hydraulic fluid shall be cause for rejection of the grader by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that graders previously accepted are similarly defective unless evidence satisfactory to the contracting officer, is furnished by the contractor that they are not similarly defective.

4.5.2.2 Sound level. The contractor shall perform sound level tests in accordance with SAE J88 and SAE J919. Certified prior test data is acceptable if available. Nonconformance to the requirements of 3.6.1 shall constitute failure of this demonstration.

4.5.2.3 Weight. The weight of the graders and grader sections shall be measured with a scale calibrated within 6 months of the demonstration. Inability to meet the minimum and maximum weight limits of 3.7.1 shall constitute failure of this demonstration.

4.5.2.4 ROPS/FOPS removal and replacement. The removal and replacement of the ROPS/FOPS canopy shall be demonstrated following procedures in the contractor's manual. Permanent deformation or damage to any part of the ROPS/FOPS canopy or its mounts, or inability to meet the requirements of 3.6.2 and 3.19 shall constitute failure of this demonstration.

4.5.2.5 Sectionalization. The type II grader less ROPS/FOPS shall be demonstrated for compliance to sectionalization requirements of 3.8. The demonstration shall be conducted on a relatively flat unimproved area of sufficient size for disassembly, maneuvering, and reassembly of the grader. Using hand tools furnished with the grader (no power tools allowed), seperate the grader into two sections. Drive the powered section under its own power for a distance of 50 feet from the remaining section and steer the powered section at least 45 degrees to the longitudinal axis of the remaining section. Drive and steer the powered section to join with the remaining section and reassemble the two sections into a composite unit using the above hand tools only. Record the time required for the maneuvering and reassembly operation. No chains, cables, hoists, blocking, or other vehicles shall be used to assist in the disassembly, maneuvering, and reassembly operations. Nonconformance to 3.8 shall constitute failure of this demonstration. The sectionalization demonstration will be compared to the disassembly and assembly instructions specified in 3.21.

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4.5.2.6 Parachute suspension provisions. The parachute suspension provisions shall be demonstrated to prove conformance to 3.19.1. Apply a force of 1.5 times the limit load as specified in MIL-STD-814 to each of the parachute suspension provisions. Weld failure or permanent deformation of the grader or suspension provisions shall constitute failure of this demonstration.

4.5.2.7 Slinging provisions. The slinging provisions shall be demonstrated to prove conformance to 3.19.2. Nonconformance to 3.19.2, weld failure, or permanent deformation of the grader or the slinging provisions shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

(a) Lift the grader and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the grader. Spreader bars are permitted. Determine the force and angles of application for each provision. Measure the clearance between each sling and the grader.
(b) Restrain the grader by anchoring the main frame and subject each slinging provision to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in (a) above. Hold each load for a minimum of 90 seconds.
(c) Repeat (a) and (b) above for each section of the type II grader.

4.5.2.8 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.19.3. Apply a force equal to the limit load as specified in MIL-STD-814 to each of the tiedown provisions. Weld failure or permanent deformation of the grader or the tiedown provisions shall constitute failure of this demonstration.

4.5.2.9 Extraction parachute provision. The extraction parachute provision shall be demonstrated to prove conformance to 3.19.4. Apply a force equal to 1.5 times the limit load specified in 3.19.4 to the provision. Weld failure or permanent deformation of the grader or the extraction parachute provision shall constitute failure of this demonstration.

4.5.2.10 Construction task. The grader shall be demonstrated as follows for a total of 40 hours to prove conformance to 3.9 and 3.10.3.

(a) Demonstrate for 33 hours the grader's ability to perform rough and fine grading operations (to include V-ditching).
(b) Demonstrate for 2 hours the grader's ability to perform high bank cutting and sloping operations.
(c) Demonstrate for 1 hour the grader's ability to grade in reverse.
(d) In the most severe soil conditions available at the demonstration site, demonstrate for 4 hours the grader's scarifying ability.

Nonconformance to 3.9 and 3.10.3 or any malfunction or breakage of mechanical or electrical components shall constitute failure of this demonstration.

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4.5.2.11 Blade pull. Blade pull tests shall be conducted in accordance with SAE J872 to prove conformance to 3.9.3. Nonconformance to 3.9.3 shall constitute failure of this demonstration.

4.5.2.12 Airdrop. The grader shall be subjected to simulated airdrop impact demonstration in accordance with MIL-STD-669 to prove conformance to 3.19. This demonstration will be conducted by the Government at a Government facility, however, acceptance of the preproduction models will be dependent on the successful completion of this demonstration. The demonstration shall consist of the following:

(a) Rig the machine in accordance with MIL-STD-669 and confirm that the rigged weight and dimensions conform to the requirements of AFSC Design Handbook DH1-11.
(b) Drop test the machine in accordance with MIL-STD-669.
(c) After the first drop, the machine shall be de-rigged, examined, started, and driven forward and backward a few feet, and all controls shall be actuated.
(d) The machine shall then be sectionalized and reassembled in accordance with the contractor's instructions to ascertain that no damage was done to the sectionalization features by the test drop. (This step is omitted for the type I machine).
(e) Re-rig and drop test the machine a second time in accordance with MIL-STD-669.
(f) After de-rigging, the grader shall be demonstrated for 8 hours performing 20 percent of each of the construction tasks in 4.5.2.10.

Failure to meet the weight and dimensional requirements of Design Handbook DH1-11, weld failure, leakage, permanent deformatin of any structural member or component, inability to reassemble the components removed for airdrop, inability to disassemble and reassemble the type II grader, or inability of the grader to perform its intended functions shall constitute failure of this demonstration.

4.5.2.13 Certification. The contractor shall certify in writing to the contracting officer that the grader complies with the environmental condition requirements of 3.9.1, the stability requirements of 3.9.4, the fording requirements of 3.9.5, the engine requirements of 3.10, the horsepower requirements of 3.10.1, the steering requirements of 3.13.1, the braking system requirements of 3.13.2, and the rail shipment requirements of 3.19. If certification is not supplied, these requirements shall be demonstrated. Nonconformance shall constitute failure of these demonstrations.

4.5.2.14 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the ROPS/FOPS requirements of 3.6.2, grader travel speed requirements of 3.9.2, and electromagnetic radiation requirements of 3.16.4. If certified test reports are not available, these requirements shall be demonstrated. Nonconformance shall constitute failure of these demonstrations.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with MIL-G-762 (for level A or B) and MIL-STD-1188 (for commercial).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each complete grader, components and attachments shall be preserved in accordance with the level A preservation and packaging requirements of MIL-G-762.

5.1.2 Level B. Each complete grader, components and attachments shall be preserved in accordance with the level B preservation and packaging requirements of MIL-G-762.

5.1.3 Commercial. Each complete grader and all items issued with the grader shall be suitably preserved and packed in accordance with MIL-STD-1188 for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent antifreeze, conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the grader in operation.

5.2. Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete grader, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-G-762.

5.2.2 Level B. Each complete grader, preserved as specified in 5.1, shall be packed in accordance with level B packing requirements of MIL-G-762.

5.2.3 Commercial. Each complete grader, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188. Commercial packing must protect items against physical and environmental damage during shipment, handling and storage.

5.3 Marking.

5.3.1 Military. Marking for level A or B protection shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The grader covered by this specification is intended for use in military construction such as rough and fine grading, low and high bank sloping, flat and V-ditching, scarifying, and general construction and maintenance of roads and air delivery in support of airborne/airmobile operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type grader required (see 1.2).
(c) Time frame required for submission of the preproduction models (see 3.3).
(d) Technical publications required (see 3.26).
(e) Time frame for completing and submitting a technical information package (see 3.29).
(f) When basic issue items are to be furnished (see 3.31).
(g) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.30.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

Custodian:
Army - ME

Review activity:
Army -AT

User activity:
Army - CE

Preparing activity:
Army - ME

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-G-52484B(ME) Grader, Road, Motorized, DED, Sectionalized and Non-sectionalized

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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